

How the Proof Works

One smooth three-dimensional incompressible flow can contain a vortex, stretch it, and drive its core toward finer scales. The global danger is whether that flow can sharpen one of its own structures so far that smooth continuation fails in finite time. Fix one velocity u , one pressure p , and one viscosity $\nu > 0$, and follow only that history:

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \nabla \cdot u = 0, \quad u(0) = u_0.$$

The transport term moves and deforms the velocity already in the fluid. Pressure responds across the whole field so that this motion remains incompressible. Viscosity diffuses a sharp change into its surroundings.

Take one vortex tube inside that field. When the surrounding strain pulls the tube along its axis, incompressibility forces a transverse contraction: a longer piece of fluid cannot keep its old cross-section without gaining volume. The core narrows. At the level of vorticity $\omega = \nabla \times u$, this motion is visible in

$$\partial_t \omega + (u \cdot \nabla)\omega = (\omega \cdot \nabla)u + \nu \Delta \omega.$$

The stretching term can amplify rotation aligned with the strain. As the core contracts, the velocity turns across a shorter transverse distance and its gradient steepens. Nonlinear motion can sharpen the vortex in this way, while $\nu \Delta \omega$ spreads the concentrated vorticity and smooths that gradient. The narrower the core becomes, the stronger both effects can become.

A narrowing vortex also changes the length and time scales on which the equation is being read. If (u, p) is viewed a factor λ deeper in space, then

$$u_\lambda(x, t) = \lambda u(\lambda x, \lambda^2 t), \quad p_\lambda(x, t) = \lambda^2 p(\lambda x, \lambda^2 t).$$

Space has contracted by λ^{-1} , the clock by λ^{-2} , and the velocity amplitude has grown by λ . Every term in the equation then carries a factor λ^3 . This is what Navier–Stokes scaling says: it compares solutions seen at different resolutions. It does not say that an evolving vortex must pass through a chain of rescaled copies.

The spatial Sobolev norms record what that comparison does to regularity:

$$\|u_\lambda(\cdot, t)\|_{\dot{H}^s} = \lambda^{s-1/2} \|u(\cdot, \lambda^2 t)\|_{\dot{H}^s}.$$

Below $s = \frac{1}{2}$, finer concentration lowers the scaled norm. Above $s = \frac{1}{2}$, it raises it. At $s = \frac{1}{2}$, it leaves the scaled size unchanged. Thus $\dot{H}^{1/2}$ is scaling-critical; it is not a conserved quantity. Saying that nonlinear transport and viscosity have equal scaling weight means only that rescaling cannot make one term disappear beside the other. It gives no cancellation, sign, or bound. Kinetic energy behaves differently:

$$\|u_\lambda(\cdot, t)\|_2^2 = \lambda^{-1} \|u(\cdot, \lambda^2 t)\|_2^2.$$

The idealized core can move to finer scales while its energy remains finite and its higher gradients grow.

Now suppose this sharpening repeats by a fixed spatial factor $\lambda > 1$. After j contractions, the scaling above gives a core length and a parabolic response time

$$\ell_j = \lambda^{-j} \ell_0, \quad \tau_j = \lambda^{-2j} \tau_0, \quad \sum_{j=0}^{\infty} \tau_j = \frac{\tau_0}{1 - \lambda^{-2}} < \infty.$$

Infinitely many scale changes can fit inside a finite T_* . The resulting Zeno cascade is a picture of one finite-energy fluid reaching arbitrarily small scales while its gradients become arbitrarily large, not a theorem that a vortex actually follows this path.

In this picture, in order for a singularity to happen, velocity has to blow up. In order for that to happen, its acceleration would also have to blow up, which means its jerk has to blow up, and so on, and so on:

$$u, \quad \partial_t u, \quad \partial_t^2 u, \quad \dots$$

The hypothetical cascade cannot stop at any fixed temporal order.

The equation carries this repeated temporal climb into its nonlinear and viscous structure:

$$\begin{aligned} \partial_t u &= \nu \Delta u - (u \cdot \nabla) u - \nabla p, \\ \partial_t^2 u &= \nu \Delta \partial_t u - (\partial_t u \cdot \nabla) u - (u \cdot \nabla) \partial_t u - \nabla \partial_t p, \\ \partial_t^3 u &= \nu \Delta \partial_t^2 u - (\partial_t^2 u \cdot \nabla) u - 2(\partial_t u \cdot \nabla) \partial_t u - (u \cdot \nabla) \partial_t^2 u - \nabla \partial_t^2 p. \end{aligned}$$

The Laplacian follows the rising temporal response, while the nonlinear term opens into every product allowed by the product rule. After three lines the bookkeeping is already hiding the relation, so put

$$V_r := \partial_t^r u, \quad Q_r := \partial_t^r p.$$

At order r , those product-rule copies occur with multiplicity $\binom{r}{a}$, so the expansion closes as

$$\begin{aligned} V_{r+1} + \sum_{a=0}^r \binom{r}{a} (V_a \cdot \nabla) V_{r-a} + \nabla Q_r &= \nu \Delta V_r, \\ Q_r &= R_i R_j \sum_{a=0}^r \binom{r}{a} V_{a,i} V_{r-a,j}. \end{aligned}$$

Incompressibility fixes Q_r through the Riesz transforms R_i , so the pressure row stays with the nonlinear row at every order.

If a singularity were forming, one would expect spatial regularity to erode as these temporal responses rise. Yet each next response contains $\nu \Delta V_r$, hence two spatial derivatives of the preceding response. That does not give regularity on its own. It tells us which spatial coordinates the equation exposes.

To measure the spatial order exposed by ΔV_r , let $\Lambda = (-\Delta)^{1/2}$ and set

$$E_s(t) := \frac{1}{2} \|\Lambda^s u(t)\|_2^2, \quad H(t) := E_{1/2}(t),$$

and retain the complete pressure-coupled nonlinear transfer

$$\mathcal{P}_s(t) := -\operatorname{Re} \langle \Lambda^s u, \Lambda^s ((u \cdot \nabla) u + \nabla p) \rangle_{L^2}.$$

Pairing the equation at height s gives

$$E'_s = \mathcal{P}_s - 2\nu E_{s+1}.$$

Differentiate this identity and use it again at each new height. At every finite order n ,

$$H^{(n)} = (-2\nu)^n E_{n+1/2} + \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{j+1/2}.$$

Acceleration, jerk, and the higher temporal responses have now exposed the Sobolev coordinates

$$E_{n+1/2} = \frac{1}{2} \|u\|_{H^{n+1/2}}^2.$$

If every finite exposed height reaches one terminal value u_* , with the finite-energy row retained, then for any k one can choose $n > k + 1$ and read

$$u_* \in L^2 \cap \bigcap_{n < \infty} \dot{H}^{n+1/2} = H^\infty \hookrightarrow C^\infty.$$

But exposure is not control. Differentiation does not bound $H^{(n)}$, the transfer terms, or $E_{n+1/2}$; it only keeps their exact relation visible. The singular picture predicts loss of spatial regularity as temporal order rises, and these identities alone do not prevent it.

Let $[0, T_*)$ be the maximal classical lifespan and suppose $T_* < \infty$. Fix one finite pair (N, M) and one cutoff $0 < T < T_*$. For $0 \leq n \leq N$, keep the whole transfer

$$\mathcal{B}_n := \sum_{a=0}^n \binom{n}{a} (V_a \cdot \nabla) V_{n-a} + \nabla Q_n.$$

Pairing the recurrence for V_n at height M gives

$$\begin{aligned} \frac{1}{2} \frac{d}{dt} \|\Lambda^M V_n\|_2^2 + \operatorname{Re} \langle \Lambda^M V_n, \Lambda^M \mathcal{B}_n \rangle_{L^2} \\ = \nu \operatorname{Re} \langle \Lambda^M V_n, \Lambda^M \Delta V_n \rangle_{L^2} = -\nu \|V_n\|_{\dot{H}^{M+1}}^2. \end{aligned}$$

The viscous term has a fixed sign, while every product-rule transfer and the pressure response remain coupled to it in this finite system. This identity shows what must be controlled; it does not supply the whole-terminal bound by itself. Theorem 4.6 supplies the bound for this maximal history:

$$\sup_{0 < T < T_*} \sum_{n=0}^N \left(\|V_n\|_{L^2(0, T; H^{M+1})}^2 + \|V_{n+1}\|_{L^2(0, T; H^{M-1})}^2 \right) < \infty.$$

Every term in the sum increases monotonically with the cutoff. Since one finite bound holds for all $T < T_*$, the limit $T \uparrow T_*$ places each adjacent pair on the full approach to T_* . With $V_{n+1} = \partial_t V_n$, that pair sits in the Hilbert triple

$$H^{M+1} \subset H^M \subset H^{M-1}$$

and gives

$$V_n \in C([0, T_*]; H^M).$$

The pressure formula supplies $Q_n \in L^1(0, T_*; H^M)$ and $\nabla Q_n \in L^1(0, T_*; H^{M-1})$, so the finitely many rows now form one pressure-complete rectangle

$$\mathcal{R}_{N,M}[u_0; T_*] = ((V_n)_{n=0}^N, (V_{n+1})_{n=0}^N; (Q_n)_{n=0}^N).$$

The bound may depend on N and M . No constant uniform over every derivative order is being used.

A larger rectangle retains each older coordinate as the identical distributional derivative of u or p . Restricting from (N, M) to (N', M') consequently gives

$$\rho_{N',M'}^{N,M} \mathcal{R}_{N,M}[u_0; T_*] = \mathcal{R}_{N',M'}[u_0; T_*].$$

The restriction maps act between the finite coordinate spaces $\mathcal{X}_{N,M}$ determined by the three rows already displayed.

$$\mathcal{X}_\infty := \varprojlim_{N,M} \mathcal{X}_{N,M}, \quad \mathcal{I}[u_0] := (\mathcal{R}_{N,M}[u_0; T_*])_{N,M \in \mathbb{N}_0} \in \mathcal{X}_\infty, \quad u \in \bigcap_{r,m < \infty} H^r(0, T_*; H^m(\mathbb{R}^3)).$$

There is no all-order estimate hidden in this notation. Each requested coordinate is obtained at a finite stage, and compatibility makes all of those finite coordinates parts of one history.

Take the zeroth temporal row after this intersection has been formed. For every finite m ,

$$u \in L^2(0, T_*; H^{m+1}), \quad u_t \in L^2(0, T_*; H^{m-1}), \quad u \in C([0, T_*]; H^m).$$

The endpoint values at different heights are compatible traces of one field. They produce a common endpoint

$$u_* := u(T_*) \in \bigcap_{m < \infty} H^m(\mathbb{R}^3) = H^\infty(\mathbb{R}^3) \hookrightarrow C^\infty(\mathbb{R}^3).$$

At T_* , set $V_0^* := u_*$. The pressure formula and binomial recurrence then define Q_r^* and V_{r+1}^* through every finite order. Smooth local existence restarts the flow from u_* . Uniqueness glues that continuation to the history already reaching T_* , contradicting finite maximality. Hence

$$T_* = \infty.$$

Only after $\mathcal{I}[u_0]$ is present do the familiar finite estimates become available as coordinate projections. Fix any finite T , a temporal order r , a spatial derivative order k , and $2 \leq p \leq \infty$, then choose

$$m > k + \frac{3}{2} - \frac{3}{p}.$$

The corresponding row gives

$$V_r \in C([0, T]; H^m) \hookrightarrow L^\infty(0, T; W^{k,p}).$$

For instance, the $m = 3$ coordinate yields

$$\int_0^T \|\nabla u(t)\|_\infty dt \leq T \|u\|_{L^\infty(0, T; W^{1, \infty})} < \infty.$$

This estimate appears because the fluid already occupies the compatible intersection; it is a finite readout of that object, not the source of it. Its source remains visible in the full dependence

$$\begin{aligned} u_0 &\longrightarrow \underbrace{\{(V_r, Q_r)\}_{r \geq 0}}_{\text{pressure-complete mixed jet}} \longrightarrow \underbrace{\left\{ \begin{array}{l} V_n \in L^2(0, T_*; H^{M+1}), \\ V_{n+1} \in L^2(0, T_*; H^{M-1}) \end{array} \right\}_{\substack{0 \leq n \leq N \\ N, M < \infty}}}_{\text{whole-terminal adjacent rows}} \\ &\longrightarrow \underbrace{\{\mathcal{R}_{N, M}[u_0; T_*]\}_{N, M < \infty}}_{\text{finite compatible rectangles}} \longrightarrow \underbrace{\mathcal{I}[u_0]}_{\text{projective intersection}} \longrightarrow \underbrace{(u_* \in H^\infty, \{(V_r^*, Q_r^*)\}_{r \geq 0})}_{\text{common endpoint and pressure-complete endpoint jet}} \\ &\longrightarrow \text{restart and uniqueness} \longrightarrow T_* = \infty. \end{aligned}$$